

EXPLAINED EXAMPLE 2 – DETERMINING PHASE

① Water is a commonly-used fluid for heat transfer applications. Willy Wonka wants to use it for both heating and cooling in the chocolate factory because he's found a secret well outside his factory and he doesn't want to push it with regulators and use a more toxic fluid – he's been busted for too many OSHA violations. Answer the questions below to help Willy Wonka design a unique water system for the factory. Hint – keep track of the states as you go through because you'll draw the whole thing on a T-v diagram at the end.

②

- a) Water comes out of the secret well at 10 MPa and 360 K (State 1). What is the *phase* of the water and what is its *internal energy*? Identify the *table* you used to answer this question.
- b) Wonka first uses the water to prepare chocolate for molding – through this machine, the process is isobaric but the temperature of the water drops to 300 K (State 2). What is the *change in internal energy* during this process?
- c) The next machine heats the water in an isobaric process until the mass-specific volume is $0.01 \text{ m}^3/\text{kg}$ (State 3) so he can use it to temper chocolate. What is the *phase* and *temperature* of the water? What is the *internal energy* at this state? Indicate which table you used to answer this question.
- d) For the final process, Wonka needs saturated vapor water (State 4) for a new type of bubble gum. How much does he have to change the *internal energy* to achieve this state? Indicate which table you used to answer this question.
- e) Draw these states and processes on a T-v diagram, labeling all states.

Given : $P_1 = 10 \text{ MPa}$

① $T_1 = 360 \text{ K}$

$P_2 = P_1$

$T_2 = 300 \text{ K}$

$P_3 = P_2$

$V_3 = 0.01 \text{ m}^3/\text{kg}$

$x_4 = 1$

Find : u_1 , phase 1

② $u_2 - u_1$

T_3 , phase 3

u_3

$u_4 - u_3$

Assume : S.C.S.

quasi-steady

a) $P_1 = 10 \text{ MPa} \rightarrow \text{Table D.2, @ } P_1 = 10 \text{ MPa}$

① $T_1 = 360 \text{ K}$

$T < T_{\text{sat}} \rightarrow \boxed{\text{liquid}}$

Table D.4B $\rightarrow \boxed{u_1 = 361.27 \frac{\text{kJ}}{\text{kg}}}$

b) $P_2 = P_1 = 10 \text{ MPa}$

② $T_2 = 300 \text{ K} \rightarrow \text{Table D.4B} \rightarrow \text{liquid}$

$$u_2 = 111.74 \frac{\text{kJ}}{\text{kg}} \Rightarrow \boxed{u_2 - u_1 = -249.53 \frac{\text{kJ}}{\text{kg}}}$$

c) $P_3 = P_2 = 10 \text{ MPa}$

$\rightarrow \text{Table D.2} \rightarrow v_f < v_3 < v_g \Rightarrow \boxed{\text{mixture}}$

① $v_3 = 0.01 \text{ m}^3/\text{kg}$

$$\rightarrow x_3 = \frac{v - v_f}{v_g - v_f} = \frac{0.01 - 0.0014526}{0.018030 - 0.0014526} = 0.52$$

$$\rightarrow u_3 = x_3 u_g + (1 - x_3) u_f = (0.48)(1393.5) + (0.52)(2545.2) \Rightarrow \boxed{u_3 = 1992.4 \frac{\text{kJ}}{\text{kg}}}$$

d) $P_4 = P_3 \rightarrow \text{Table D.2} \rightarrow u_4 = u_g = 2545.2 \frac{\text{kJ}}{\text{kg}}$

② $x_4 = 1$

$$\boxed{u_4 - u_3 = 552.8 \frac{\text{kJ}}{\text{kg}}}$$

